

TEST PLAN

Project / Product: VWO (Visual Website Optimizer) - Login Dashboard **Release / Version:** [Release Version]
Document Version: 1.0 **Prepared By:** [QA Lead Name] **Reviewed By:** [Names / Roles] **Approved By:** Seema
Planned Test Start: [Date] **Planned Test Completion:** [Date] **Target Release Date:** [Date] **Status:** Approved

1. Document & Release Overview

1.1 Purpose

This Test Plan defines the quality assurance strategy, scope, test approach, risks, and release criteria for validating the **VWO Login Dashboard** described in JIRA ticket **SCRUM-1** ("Login page to VMO.com website", Priority: High, Status: To Do).

The objective is to establish a shared understanding among QA, Engineering, Product, and Business stakeholders of how the login experience will be validated - and which requirements are still missing or ambiguous enough to block that validation.

1.2 Release Overview

Per the PRD embedded in SCRUM-1, this release delivers a secure, intuitive login experience at `app.vwo.com` that connects users to VWO's experimentation, personalization, and analytics platform.

Scope highlights from the PRD:

- Email + password authentication with secure session management
- Remember Me persistent-session option
- Forgot Password / password reset flow
- Sign in with Google (social login)
- Sign in using SSO (enterprise)
- Sign in with Passkey
- Light and Dark mode theming
- Responsive / mobile-optimized design
- WCAG 2.1 AA accessibility
- Rate limiting against brute force
- Analytics integration for login success/failure tracking
- Free-trial signup CTA for new users

Not confirmed: which of the above are in this release vs. deferred to Phase 2 / Phase 3 (see Gaps & Questions).

1.3 References

- | Ref | Document |
 - | R1 | JIRA SCRUM-1 ticket (PRD embedded in description) |
 - | R2 | Ticket attachment: `image-20260812-132015.png` (login page mockup) |
 - | R3 | Product Requirements Document.docx (repo root) |
 - | R4 | VMO platform overview - `.commandcode/skills/testPlan_gen/assets/vmo_details.md` |
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2. Objectives & Scope

2.1 Test Objectives

- Validate the login form and all sign-in paths work per the PRD and the attached mockup.
- Verify failed-authentication handling: error messages, field validation, rate limiting.
- Validate account recovery (Forgot Password) end to end.
- Verify post-login handoff to the VWO dashboard for each successful path.
- Validate security controls: HTTPS, credential handling, session lifecycle.
- Validate non-functional requirements: page load $\leq 2s$, responsive layout, WCAG 2.1 AA.
- Confirm existing functionality is not regressed by the new login page.

2.2 In Scope

Functional

- Email/password login (happy path + validation + error paths)
- Remember Me behavior (session persistence)
- Forgot Password flow (email-based reset)
- Google social login
- SSO login
- Passkey login
- "Start a FREE TRIAL" CTA and "New to Wingify?" redirect
- Light/Dark mode toggle on login page

Integration

- Handoff to VWO dashboard post-login
- Analytics events on login success/failure
- Third-party identity providers (Google, SSO IdP, passkey provider)

Data

- Login form field validation (email format, required fields)
- Session/token lifecycle and timeout

Platforms

- Browsers: Chrome, Edge, Firefox, Safari (desktop)
- Mobile: iOS Safari, Android Chrome (responsive layout)

Non-Functional

- Performance: Yes (page load $\leq 2s$, concurrent login attempts)
- Security: Yes (encryption, rate limiting, session security)
- Accessibility: Yes (WCAG 2.1 AA)
- Compatibility: Yes (browser matrix above)

2.3 Out of Scope

- **Biometric / adaptive authentication** (Future Enhancements in PRD - not in a committed release)
 - **Progressive Web App** (Future Enhancements)
 - **A/B testing / personalization of the login page** (Future Enhancements)
 - Marketing-tool integrations beyond login analytics
 - Backend/API implementation details of the auth service (covered at API level only where needed for validation)
 - Performance/load testing beyond login-page load-time and brute-force throttling
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3. Product & Quality Risk Overview

3.1 Product Overview

- **Application:** VWO login dashboard at `app.vwo.com`
- **Architecture:** Web app with a login form, third-party OAuth/SSO/passkey providers, and a backend auth service; analytics events on auth outcomes.
- **User types:** Digital marketers, product managers, UX designers, developers, CRO specialists, data analysts (individual accounts); enterprise users via SSO.
- **Critical workflows:** Any successful authentication -> dashboard access; password reset; account recovery.
- **External systems:** Google identity provider, enterprise SSO IdP (SAML/OAuth), passkey/WebAuthn provider, analytics.

3.2 Critical Business Flows

- Email/password sign-in -> dashboard
- Google sign-in -> dashboard
- SSO sign-in -> dashboard
- Passkey sign-in -> dashboard
- Forgot password -> reset email -> new password -> sign-in
- Remember Me -> session persists across browser restarts

3.3 Quality Risks

Risk	Impact	Probability	Risk Level	Mitigation
SSO/Google/passkey provider integration failure	High	Medium	High	API + E2E validation of each provider
Session/Remember Me security flaw (token leakage, hijack)	High	Low	High	Security review + session lifecycle tests
Brute-force rate limiting not effective	High	Medium	High	Negative tests on throttling thresholds
Login page load > 2s on mobile/slow networks	Medium	Medium	Medium	Performance tests on standard + throttled connections
WCAG 2.1 AA gaps block keyboard/screen-reader users	Medium	Medium	Medium	Manual + automated a11y checks
Dark mode rendering defects	Low	Low	Low	Visual regression on both themes

3.4 Risk-Based Testing Prioritization

Priorities follow: business criticality (login is the entry point for every user), security exposure (auth data, session tokens), third-party integration complexity, and accessibility/compliance obligations. High-risk areas get higher test depth and earlier execution.

4. Test Strategy

4.1 Overall Testing Approach

Risk-based and layered: API/service-level validation of the auth endpoints plus E2E validation of every sign-in path in the UI, with security and performance checks on the areas of highest exposure.

4.2 Test Levels

- | Test Level | Purpose | Primary Owner |
- | Unit Testing | Validate individual components | Development |
- | API/Service Testing | Validate auth endpoints, validation, rate limiting | QA/Development |
- | Integration Testing | Validate IdP integrations (Google, SSO, passkey) | QA/Development |
- | System Testing | Validate complete login behavior | QA |
- | End-to-End Testing | Validate each sign-in path end to end | QA |
- | UAT | Validate business acceptance | Business/Product |

4.3 Test Types

- Functional, Integration, Regression, Smoke, Sanity, API, UI, Database, Exploratory, Compatibility, Performance (load-time), Security (auth-specific), Accessibility

4.4 Test Design Approach

- Positive scenarios for each sign-in path
- Negative scenarios (wrong password, unregistered email, malformed email, locked account)
- Boundary conditions (empty fields, max-length inputs, whitespace-only)
- Error handling (network failure, IdP cancellation, expired reset token)
- Role/permission combinations (individual user vs. enterprise SSO user)
- Integration failures (IdP down, analytics down)
- Exploratory testing on the login flows

4.5 Regression Strategy

Regression focuses on: core sign-in paths, session persistence, password reset, and any existing dashboard functionality reachable post-login. The regression suite is reviewed against the release impact analysis before execution.

5. Test Coverage & Traceability

5.1 Coverage Strategy

Coverage is established against the PRD functional requirements, the mockup attachment, business-critical workflows, and identified security/performance risks. Because the ticket has **no acceptance criteria**, traceability is mapped to PRD requirement statements (see 5.2); each Gaps & Questions item that stays open is treated as an untestable requirement until resolved.

5.2 Requirement Traceability

Requirement -> Test Scenario -> Test Case -> Execution Result -> Defect

Traceability will be maintained in **[Jira/ALM/Xray/Zephyr/etc.]** once the test scenarios below are approved and converted to cases.

Req (PRD section)	Scenario(s)
Login Process - email/password	S-01, S-02, S-03
Remember Me	S-04
Password Management - forgot/reset	S-05, S-06
Social Login (Google)	S-07
SSO	S-08
Passkey	S-09
Session Management / timeout	S-10
Security - rate limiting	S-11
Security - HTTPS / token handling	S-12
User Input Validation	S-02, S-03
Interface Design - responsive	S-13
Accessibility (WCAG 2.1 AA)	S-14
Branding - Light/Dark mode	S-15
Integration - dashboard handoff	S-01, S-07, S-08, S-09
Analytics integration	S-16
Free Trial CTA	S-17
Performance - page load	S-18

5.3 Coverage Expectations

- Requirement coverage (all PRD functional requirements mapped)
 - Critical workflow coverage (every sign-in path)
 - Risk coverage (security + performance + accessibility items)
 - Test execution coverage and automation coverage tracked in the test management tool
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6. Test Environment & Test Data

6.1 Test Environments

Environment	Purpose	Owner	Status
DEV	Developer validation	Engineering	[Status]
QA/SIT	Functional & integration testing	QA	[Status]
UAT	Business validation	Product/Business	[Status]
Staging	Production-like validation	Engineering/QA	[Status]

6.2 Environment Requirements

- Application version and build for the login dashboard
- Backend auth service with test-mode IdPs (Google test account, SSO test IdP, passkey test provider)
- Email service for password-reset and verification emails
- Analytics endpoint to capture login events
- Browsers/devices per the matrix in 2.2

6.3 Test Data Strategy

- Positive data: valid email/password accounts (individual), enterprise SSO test account
- Negative data: unregistered email, wrong password, malformed email, whitespace inputs
- Boundary data: max-length email/password, empty fields, 1-char password
- Role-based data: standard user vs. SSO/enterprise user
- Integration data: test Google account, test SSO IdP users
- Production-like data only where permitted and masked per privacy policy

6.4 Environment/Data Risks

- Real IdPs may not offer full test environments -> mock or use dedicated test accounts
 - Reset emails may be delayed in QA environments -> allow for delivery time in scheduling
 - No seed test data is defined in the ticket -> data set must be created and approved
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7. Test Automation Strategy

7.1 Automation Objectives

Automation targets repeatability of the highest-risk flows: every sign-in path, session persistence, rate limiting, and smoke checks - to shorten regression cycles and support release confidence.

7.2 Automation Scope

- All sign-in paths (email/password, Google, SSO, passkey) as E2E (existing repo framework: Selenium - see repo README)
- API-level auth validation and rate-limit checks
- Smoke: page loads, form renders, sign-in with a known-good account
- Session timeout / Remember Me behavior

7.3 Manual Testing Scope

- Exploratory testing of error states and edge cases
- Visual/design validation (Light/Dark mode, responsive layouts)
- Accessibility validation (screen readers, keyboard-only)
- Usability judgment calls (error-message clarity)

7.4 Automation Framework

- Framework: [Selenium/Playwright per repo convention - to be confirmed]
- Language: [to be confirmed]
- Test management integration: [Jira/Xray/Zephyr - to be confirmed]
- CI/CD integration, reporting, parallel execution, test data management: [to be confirmed]

7.5 Automation Quality

Effectiveness is evaluated via pass rate, stability/flakiness, execution time, maintenance effort, defect detection effectiveness, and useful regression coverage.

8. Defect Management

8.1 Defect Lifecycle

New -> Assigned -> In Progress -> Fixed -> Retest -> Closed (organization process applies).

8.2 Severity

| Severity | Definition |

| Critical | Login unavailable for all users, credential/session data exposure, severe security issue |

| High | One sign-in path fully broken, password reset broken, rate limiting bypassed |

| Medium | Functionality impacted with workaround (e.g., dark-mode rendering issue) |

| Low | Minor functional/cosmetic issue |

8.3 Defect Reporting

Each defect: summary, environment, build/version, preconditions, steps to reproduce, expected result, actual result, severity/priority, evidence, logs/API response.

8.4 Defect Triage

Triage conducted **[daily/weekly/as required]** with QA, Engineering, Product/Business. Critical and high defects escalated per agreed timelines.

8.5 Defect Closure Criteria

- Fix deployed to the appropriate environment
- Original issue successfully retested
- Relevant regression completed
- No related issues remain open

9. Entry, Exit & Quality Gates

9.1 Entry Criteria

- Approved requirements / resolved Gaps & Questions (see §2) - **currently blocking**
- Testable build deployed
- Environment stable, test data available
- Critical dependencies (IdPs, email service, analytics) available
- Build smoke testing successful

9.2 Exit Criteria

- Planned critical/high-priority testing completed (all sign-in paths)

- Critical business workflows passed
- Required regression completed
- No unresolved Critical defects (unless explicitly accepted)
- High-severity defects within agreed risk threshold
- Non-functional testing (perf, security, a11y) completed
- Test results and risks communicated to stakeholders

9.3 Quality Gates

Gate 1 - Build Acceptance: Build deployed, smoke tests passed, no deployment blockers. **Gate 2 - Functional Test Completion:** Critical functionality validated, major defects triaged. **Gate 3 - Regression Completion:** Planned regression executed, results reviewed, outstanding risks assessed. **Gate 4 - Release Readiness:** Exit criteria met, remaining risks documented, business acceptance obtained, QA release recommendation issued.

10. Roles, Responsibilities & Dependencies

10.1 Roles

Activity	QA Lead	QA Engineer	Developer	Product/BA
Test Strategy	A/R	C	C	C
Test Design	A	R	C	C
Automation	A	R	C	I
Defect Triage	A/R	R	R	C
Requirement Clarification	C	C	C	A/R
Release Recommendation	A/R	C	C	C
UAT	C	C	I	A/R

10.2 Dependencies

- Development build availability
- QA environment availability
- Google/SSO/passkey test provider readiness
- Email service availability (reset flow)
- Analytics endpoint availability
- Test data approval
- Business/UAT availability

10.3 Assumptions

- Requirements are approved before test design begins (currently the PRD is narrative-only - see Gaps)
- Environments are available per schedule
- External IdPs are accessible during testing
- Defects are resolved per agreed priorities

10.4 Constraints

- No acceptance criteria in the ticket
- IdP test environments may be limited

- Reset-email delivery delays in QA envs
- Unknown release window

Mitigations: see Gaps & Questions and 6.4.

11. Metrics, Reporting & Deliverables

11.1 QA Metrics

Test execution progress, pass/fail/blocked %, requirement coverage, risk coverage, defect distribution by severity, defect aging, defect reopen rate, regression pass rate, automation coverage, automation stability, production defect leakage.

11.2 Reporting

Daily/periodic test status, defect status, risk status, test execution dashboard, release readiness status, final test summary.

11.3 Test Deliverables

Test Plan, test scenarios/cases, automation suite, test data, defect reports, test execution results, QA status reports, Test Summary Report, release recommendation.

12. Release Readiness & Post-Release Validation

12.1 Release Readiness

QA recommendation based on test execution results, defect status, risk assessment, critical workflow status, regression results, non-functional results, exit criteria, outstanding known issues.

12.2 Release Decision

GO - all critical quality gates satisfied, remaining risks acceptable. **GO WITH ACCEPTED RISKS** - known issues reviewed and accepted by business/product. **NO-GO** - one or more critical gates not satisfied or unacceptable risk remains.

12.3 Production Validation

Production smoke: login page loads, email/password + at least one third-party path (Google) sign-in, dashboard handoff, error/log monitoring, integration health checks.

12.4 Post-Release Review

Review production defects, escaped defects, coverage gaps, automation gaps, environment issues, process issues, lessons learned; feed improvements into the next release.

Test Scenarios (P0/P1/P2)

| ID | Priority | Trace | Scenario | Precondition | Expected Result |

| S-01 | P0 | PRD Login Process | Email + password with valid credentials signs in and lands on the VWO dashboard | Valid account exists | User authenticated, redirected to dashboard, success state visible |

| S-02 | P0 | PRD Input Validation | Field validation: empty, malformed, whitespace-only email/password | On login form | Inline error per field, no submit of invalid data |

| S-03 | P0 | PRD Login Process | Invalid credentials (wrong password / unregistered email) | Login form | Clear actionable error, no session created |

| S-04 | P0 | PRD Remember Me | Remember Me checked -> session persists after browser restart; unchecked -> session expires | Valid account | Session behavior matches the checkbox state |

| S-05 | P0 | PRD Forgot Password | Forgot Password -> reset email -> set new password -> sign in with new password | Registered email | Full reset flow succeeds; old password no longer works |

| S-06 | P1 | PRD Password Recovery | Expired/used reset token handled gracefully | Reset token issued | Clear error; user can request a new link |

| S-07 | P0 | PRD Social Login | Sign in with Google succeeds -> dashboard; cancel -> returns to login | Test Google account | Successful auth or clean cancel, no partial state |

| S-08 | P1 | PRD SSO | Sign in using SSO (enterprise account) succeeds -> dashboard | Test SSO IdP account | Enterprise user authenticated and redirected |

| S-09 | P1 | PRD Passkey | Sign in with Passkey succeeds; passkey unavailable path handled | Passkey test device | Passkey auth works or graceful fallback message |

| S-10 | P1 | PRD Session Mgmt | Session timeout: idle session expires per configured timeout | Logged-in session | Session ends at timeout; re-login required |

| S-11 | P0 | PRD Rate Limiting | Repeated failed attempts trigger rate limiting; legitimate user not blocked | Login form | Throttling activates after N failures; valid user can still log in |

| S-12 | P1 | PRD Security | HTTPS enforced on login; no credentials in URLs/logs | Browser devtools | All login traffic over HTTPS, no credential leakage |

| S-13 | P1 | PRD Responsive | Login renders correctly on mobile and tablet viewports | Browser at mobile widths | No overlap/clipping; touch targets usable |

| S-14 | P1 | PRD A11y | WCAG 2.1 AA: keyboard-only navigation, screen reader labels, focus states | Login page | Full keyboard operability, labeled inputs, visible focus |

| S-15 | P2 | PRD Theme | Light and Dark mode both render correctly with VWO branding | Theme toggle | Both themes consistent, no contrast issues |

| S-16 | P1 | PRD Analytics | Login success/failure analytics events fire correctly | Analytics endpoint | Events recorded for success and failure outcomes |

| S-17 | P2 | PRD Free Trial CTA | "Start a FREE TRIAL" navigates to the signup/trial flow | Login page | Correct destination page opens |

| S-18 | P1 | PRD Perf | Login page loads within 2s on standard and throttled connections | Clean session | Load ≤ 2s on standard connection; degradation measured on throttled |

Gaps & Questions for the author

These are the findings from the requirement gap-analysis (■ present / ■■ ambiguous / ■ missing). They block or constrain test design - please resolve before sign-off.

■ Missing / blocking

1. **No acceptance criteria.** The PRD is narrative. There is no Given/When/Then or observable pass/fail for any requirement. Convert the PRD bullets into testable ACs, or confirm each scenario in \$Test Scenarios.
2. **Scope is not defined for this release.** SSO, Passkey, Google, 2FA, Remember Me, dark mode, analytics - the PRD lists all as requirements but does not say which ship in this release vs. Phase 2/3. Which sign-in paths are actually in scope?

3. **"VMO.com" vs "VWO"** - the ticket summary says "VMO.com website" while the PRD and mockup say VWO (app.vwo.com). Which host is the real target? A typo here could point tests at the wrong environment.
4. **Test data & environments not defined.** No seed users, no test accounts for Google/SSO/passkey, no environment list. Confirm what QA can use, especially for third-party IdPs.
5. **2FA/MFA** is listed as "Optional 2FA support" - is it in scope? If yes, which methods (TOTP, SMS, authenticator app)?
6. **Session timeout value** is "configurable" but not specified. What is the actual timeout for this release?

■ ■ Ambiguous

1. **Rate limiting thresholds** - "protection against brute force through request throttling" with no numbers. How many failed attempts before lockout, and for how long?
2. **Password complexity rules** - "enforced security standards" but no minimum length, complexity, or expiry policy.
3. **Remember Me semantics** - persistent session for how long? Any security trade-off (e.g., token rotation) expected?
4. **"Seamless transition to main dashboard"** - what state carries over (recent activity, context preservation)?
5. **Error message inventory** - none of the exact messages are specified ("clear, actionable" only). Provide the expected messages for each failure type so tests can assert exact text.
6. **Analytics integration** - which events, names, and payloads are expected for login success/failure?
7. **Passkey/SSO/Google fallback behavior** - if a provider is unavailable, what should the user see?

■ Present (usable as-is)

- Login form fields and layout (from mockup attachment)
- Page load $\leq 2s$ target
- WCAG 2.1 AA compliance target
- HTTPS enforcement
- Light/Dark mode theming
- Responsive design requirement
- Free-trial CTA flow

Risks & Assumptions

- **Assumption:** the attached mockup (image-20260812-132015.png) reflects the intended final UI - to be confirmed against the actual build.
- **Assumption:** Google, SSO, and passkey providers offer test environments usable by QA.
- **Risk:** without acceptance criteria, test pass/fail is subjective - QA sign-off may be contested.
- **Risk:** third-party IdP instability can block sign-in path testing.
- **Risk:** if "VMO" vs "VWO" is unresolved, tests may run against the wrong host.

Entry / Exit criteria

(Consolidated from §9.)

Entry: approved requirements + resolved gaps; testable build; stable environment; test data; critical dependencies available; smoke passed. **Exit:** P0 scenarios passed; critical workflows validated; regression complete; no unresolved Critical defects; high-severity within threshold; non-functional checks done; results communicated.

--- HUMAN REVIEW GATE ---

Status: APPROVED on 2026-08-19 by Seema.

The open questions below remain open for the ticket author and should be resolved before test case design begins, but they no longer block this test plan.

Assumptions made:

- I assumed the mockup attachment is the intended UI for this release.
- I assumed the repo's existing Selenium-based framework (per repo commits) is the automation base - to be confirmed.
- I assumed Google/SSO/passkey providers provide QA test environments.
- I assumed this release is scoped to Phase 1 (Core Authentication) plus the visible mockup elements, since the ticket does not state scope.

Open questions (carry forward to test case design):

1. Which sign-in paths are in scope for this release (email/password, Google, SSO, Passkey - all? some)?
2. Is there an updated/authoritative version of the acceptance criteria, or should the scenarios above be treated as the ACs pending your review?
3. Is the target host `app.vwo.com` (ticket title says "VMO.com" - typo?).
4. What test accounts and environments can QA use (esp. Google/SSO/passkey)?
5. What are the rate-limiting and password-complexity rules for this release?
6. What is the configured session timeout and Remember Me duration?